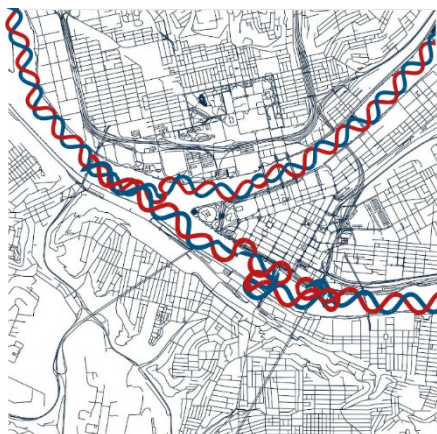




New Frontiers in Nucleic Acids

Wednesday, September 16, 2026

Mellon Institute, Carnegie Mellon University,
Pittsburgh, PA

Registration and Breakfast: 8:30am – 8:55am

Symposium runs 9:00 am – 5:00 pm with break for Lunch

Keynote Talks

Speaker	Affiliation	Talk Title
Prof. Naoki Sugimoto	Konan University Frontier Institute for Biomolecular Engineering Research (FIBER)	“To B or not to B” in Nucleic Acids Chemistry
Prof. Alex Deiters	University of Pittsburgh	Covalent Aptamers

Invited Talks

Speaker	Affiliation	Talk Title
Prof. Danith Ly	Carnegie Mellon University	GammaPNAs: As Orthogonal Nucleic Acid Recognition Codes for Organizing Molecular Self-Assembly and for Molecular Computing

Speaker	Affiliation	Talk Title
Dr. David Heisler	Duquesne University	G-Quadruplex and i-Motif Structures in the 5'UTR of DNA Methylation Genes Modulate Expression
Dr. Anna M. Kietrys	Pittsburgh, PA	Ribozyme circRNA: An Evolutionary Artefact or a Conserved Feature?
Prof. Rebecca Taylor	Carnegie Mellon University	Condensates, encapsulations and lattice materials with some help from peptide nucleic acid and generative design
Dr. Jonathan Henninger	Carnegie Mellon University	RNA and emergent regulation in gene control

12-Minute Student Talks

Speaker	Affiliation	Talk Title
Ruwani Madushika Dalath	Carnegie Mellon University	γ PNA-based RNA imaging reveals a role of APBs in localizing TERRA to telomeres in ALT cancer cells
Keito Obata	Kyoto University	Development of intracellular RNA-G4 sensor

Speaker	Affiliation	Talk Title
Alec Johnson	Carnegie Mellon University	Trace gene and mRNA sensing by PNA-mediated electrophoresis
Ayumi Kikkawa	Kyoto University	Development of a Peptide–ssDNA-Based Coacervate System for mRNA Delivery
Chris Blum	Carnegie Mellon University	EZON Modification of Extracellular Vesicles for Programmable and Targeted Delivery
Megumi Kiyokawa	Kyoto University	Coacervates for cytosolic infusion of antibodies: design and physicochemical study
Shubhankar Kundu	University of Pittsburgh	A DNA based approach to pattern proteins on surface

Poster Session 4:00 am – 5:00 pm